# Alex José Brinson

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https://alexjbrinson.github.io/

## EDUCATION

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| Sept 2014 – June 2018 | **California Institute of Technology, Pasadena, CA**<br>BS in Physics ; GPA: 3.6 |
| Aug 2010 – May 2014 | **Cedar Ridge High School, Round Rock, Texas**<br>GPA: 4.00/4.00 (Unweighted), 5.69/6.00 (Weighted) |

## TECHNICAL SKILLS

**Languages and Software:**

- Python, Mathematica, C++, C, MATLAB, R, LaTeX, kramdown, Microsoft Office

**Hardware:**

- Experience with Oscilloscopes, Interferometers, Spectrometers, Monochromators, Magnetometers, Soldering

## RESEARCH EXPERIENCE

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| Jun –Aug 2018 | **Cosmic DAWN Summer Research Fellow** 40 hours/week<br>Investigated method to infer galaxy redshifts based on angular correlation functions from large samples COSMOS catalogue data. |
| Jan –Jun 2018 | **nEDM Group Undergraduate Researcher** 9 hours/week<br>Adapted Mathematica FCS simulation to model the peculiar geometry of the Oak Ridge nEDM apparatus. |
| Jun – Aug 2017 | **nEDM Group Research Fellow** 40 hours/week<br>Created Mathematica simulation of Field Compensation System (FCS) to optimally offset ambient magnetic fields. Performed thorough analysis of lab magnetization sources to determine a suitable experiment location. |
| Oct – Dec 2016 | **Keck Array Undergraduate Researcher** 9 hours/week<br>Calibrated Cernox thermometers, analyzed load curves in Python to determine Power vs. Temperature behavior. |
| Jun – Aug 2016 | **Keck Array Research Fellow** 40 hours/week<br>Characterized dark properties of TES bolometers for later use in CMB detection. Assisted with cryostat assembly. |

## OTHER WORK EXPERIENCE

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| Apr – Jun 2017 | **Ph 7 Undergraduate Teaching Assistant** 9 hours/week<br>Graded pre-lab exercises, assisted with in-lab procedures and guided post-lab analyses. |
| Apr 2015 – Mar 2017 | **Student Waiter** 6 hours/week<br>Set up and cleaned up dinners for undergraduate students. |

## HONORS & AWARDS

- 2017 CEU (Conference Experience for Undergraduates) Award Recipient
- 2016 Edward C. and Alice Stone SURF Fellow